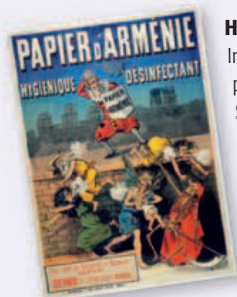


BEFORE

Scientific discoveries and improved sanitation in the 19th century laid the foundations for the medical revolution of the 20th.

SANITATION

In British cities c. 1830, **infant mortality** was about 50 percent—half of all children born died before 5 years of age. Most deaths were caused by water contaminated with sewage. In the second half of the 19th century, sewers were built and water was piped into cities. By 1914, infant mortality was less than 10 percent.



1890S DISINFECTANT ADVERTISEMENT

patients. His discovery led to better hospital hygiene. Carbolic acid was used to sterilize surgical equipment and wounds from the 1860s.

BACTERIA AND DISEASE

In the 1870s and 1880s, the German physician Robert Koch isolated the different bacteria that cause tuberculosis, cholera, and anthrax, allowing more effective **vaccines** << 327 to be developed.

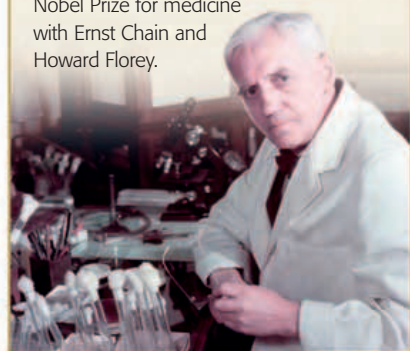
VITAMINS

In the early 1900s, scientists investigating the link between **diet and health** identified key nutrients in food. In 1912, **Cashmir Funk** named them “vitamines”; “vita” meaning life, and “amine” from compounds isolated in his experiments.

BRITISH SCIENTIST (1881–1955)

ALEXANDER FLEMING

Born in Scotland, the pharmacologist and biologist Alexander Fleming made huge contributions to the fields of bacteriology, immunology, and chemotherapy. He is famous for isolating the antibiotic substance penicillin from the fungus *Penicillium notatum*. He shared the 1945 Nobel Prize for medicine with Ernst Chain and Howard Florey.



Protecting the troops

Sailors in the British Royal Navy receive inoculations against cholera before deployment overseas during World War I.



World Health

The past century has produced astonishing advances in health and medicine. Some infectious illnesses have been eradicated, but there are still many killer diseases at large, especially in the developing world.

In the first few decades of the 20th century, Dr. Sara Josephine Baker pioneered the education of women in the basic hygiene and dietary care of their children in the slums of New York City. Baker's health education program led to a significant reduction in infant mortality rates, and by the 1950s most industrialized countries had adopted similar measures.

Health education, better sanitation, and rising living standards in industrialized countries combined to produce steady, if unspectacular, improvements in public health. However, some developments were more dramatic. In 1928, Alexander Fleming (see left) found that a rare strain of mold inhibited the growth of bacteria. This discovery enabled him to develop penicillin—an infection-fighting “antibiotic” drug. Mass production of penicillin began in 1945. By the 1950s, antibiotics were providing effective treatments for many terrible diseases, including syphilis, gangrene, and tuberculosis (TB).

In the second half of the 20th century, mass vaccination programs, funded by governments and the World Health Organization (WHO), further reduced

mortality rates. In 1980, the WHO announced the global elimination of smallpox. Polio, too, is close to being eradicated; in fact, the general trend for most infectious diseases is downward.

Furthermore, chemotherapy has made huge strides in the fight against cancer, and heart and organ transplants are now commonplace. Perhaps most amazingly, in vitro fertilization (IVF) can allow couples who would previously have been childless to have a family.

Rich and poor

Unfortunately, the great achievements in health care and medicine are not always passed on to the poor. The diphtheria, tetanus, whooping cough (DTP) vaccine has been available since the 1940s, and has largely eradicated these diseases in the developed world.

However, in Africa, India, and Eastern Asia, less than 50 percent of children are covered by DTP programs. WHO

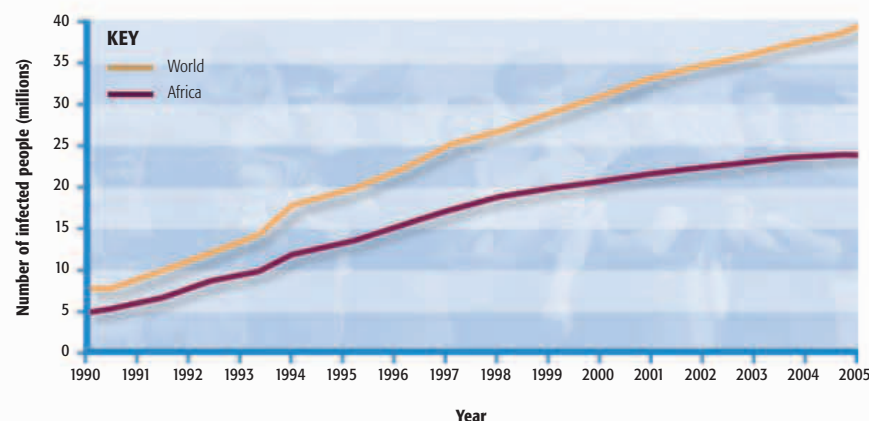


Changing attitudes

A 1950s ad (left) shows Hollywood star and future US president Ronald Reagan promoting cigarettes. Two decades later, a French cancer charity (right) warns of the health risks of smoking to pregnant women and their babies.

statistics show that children in the poorest countries have a mortality rate 2.5 times higher than their counterparts in developed nations.

The greatest killers in the developing world are waterborne illnesses such as typhoid, cholera, dysentery, and even diarrhea. In fact, nearly 40,000 people die each day from drinking water contaminated by sewage. Providing sanitation and clean drinking water is a major challenge for the governments of developing countries. Respiratory diseases such as bronchitis, emphysema, pneumonia, and lung cancer also take a heavy toll. These diseases are often linked to the inhalation of smoke from indoor cooking fires. A simple solution is to provide a smoke hood and a chimney, rather than an open



Global trends in HIV infection

Around 40 million people worldwide (nearly two-thirds of whom live in Africa) are infected with HIV, the virus that causes AIDS. Over 4 million new cases occurred in 2006 alone. “Cocktails” of antiretroviral drugs have improved health and life expectancy for people with HIV.